

Executive Business Report: Sales Forecasting and Demand Intelligence

Executive summary: The Superstore business generated \$2,261,537 across 4,922 orders. The forecasting system estimates the next three months at \$37,040, \$28,310, and \$62,861. The system is useful for planning, but should be treated as a decision-support tool because promotion, price, stockout, and holiday data were not available. The supplementary video game sales dataset was included as year-level external market context, not as a direct causal feature.

Finding	Evidence
Top category	Technology generated \$827,456.
Top region	West generated \$710,220.
Shipping	Average ship time was 3.96 days; Central was slowest at 4.07 days.
Best local model	Prophet fallback - trend/month regression had MAE \$14,681, RMSE \$18,498, and MAPE 17.7%.
Supplementary data	vgsales.csv added 16,327 rows. 3 Superstore years matched by year for context.

Forecast and Anomalies

The next three monthly forecasts are \$37,040, \$28,310, and \$62,861. Confidence ranges should be interpreted cautiously because the local fallback model does not produce formal statistical intervals. A practical business range is plus/minus the validation RMSE.

Top anomalies: 2015-03-16 had \$37,704, likely due to festive sale, promotion, or bulk buying. 2018-11-26 had \$35,999, likely due to festive sale, promotion, or bulk buying. 2018-11-12 had \$30,572, likely due to festive sale, promotion, or bulk buying.

Demand Segments and Stocking Strategy

Segment	Sub-categories	Stocking action
Growing Demand	Copiers	Increase purchase planning gradually; review monthly.
High Volatility Watchlist	Machines	Use shorter replenishment cycles; avoid heavy overstock.
High Volume, Stable Core	Accessories, Binders, Chairs, Phones, Storage, Tables	Maintain safety stock; review service levels weekly.
Low Volume / Declining Demand	Appliances, Art, Bookcases, Envelopes, Fasteners, Furnishings, Labels, Paper, Supplies	Limit bulk buys; use clearance planning where needed.

Recommendations and Limitation

1. Protect inventory for Technology, the largest category at \$827,456.
2. Review regional replenishment in Central, where shipping takes 4.07 days on average.
3. Use anomaly weeks as promotion planning signals rather than deleting them from the data.

Limitation: The system uses sales history plus weak year-level external context; adding price, discount, stockout, and promotion calendars would improve forecast reliability.

Project pain points documented: supplementary dataset has no direct join key, missing local forecasting libraries, small monthly history, no promotion/stockout features, and deployment requiring user account access.